

Darshan Kumar

Project Manager & Software Engineer

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Education

University of Louisiana at Lafayette | IEEE | CAPE | ULift | Honors College

B.S. Computer Science with concentration in Computer Engineering | Minor: Mathematics | **Senior**

Aug 2023 – May 2026

Certifications & Licenses: ITAR, HAM Radio Technician (Call Sign: KJ5MAG)

Experience

METECS Intern - Houston, TX | **ER711**, Simulation and Graphics Branch

May 2025 - Aug 2025

➤ **USSF GROOT** - *United States Space Force Generic Robotics On-Orbit Trainer*

- Investigated and developed a POC for a MIL-STD-1553 and **IRIG 106 Chapter 10** encoder for use with USSF flight systems.
- Expanded hand controller selection with three additional types of hand controllers.
- Improved UI/UX for the TRICK pre-launch interface.
- Exposure to TrickCFS, cFS, cFE, FSW, NPR 7150.2D, Hand Controllers, CHAPEA, and B16 SEAL, PIT, and High Cell.

CAPE - *Cajun Advanced Picosatellite Experiment*, Lafayette, LA | **EECE Undergraduate Student Research** at **UL Lafayette**

Aug 2023 - Present

➤ **Lead Software Developer** | Fall 2023 high-altitude weather balloon

- Leading, planning, and teaching students in developing software and systems integration.
- Developing C code on the STM32 Microcontroller for various sensors through STM32 IDE.

➤ **Project Manager & Lead Electronics and RF Engineer** of **CAPE-4** (CubeSat) & **ESG-Grid** (Experimental Smart Ground-Station Grid)

- **Patented, ITAR Compliant**, globally deployed, **RF Ground-Station solution** primarily for transmission and reception of data to and from Cube Satellites and high-altitude weather balloons.
- **Lead a successful PDR** with prepared legal documentation, scope of works, technical requirements, block diagrams, technical presentations, and software flow-charts for management, professors, and industry mentors.
- Successfully built a team of software engineers to develop ESG-Grid's ground-station infrastructure, data-handling, and web app.
- **ULift** - Significantly contributed towards a rocketry mission utilizing the ESG-Grid's ground-station to rebuild and graphically model the rocket's attitude during flight up to 10,000 ft.
- **Command & data handling, attitude determination & control system**, radio system system **design and architecture** for CubeSats.

➤ **Assistant Program Manager** | CAPE Program

- Assisting and mentoring agile development, technical development, and documentation efforts of the CAPE-Twiggs, CAPE-4, ESG-Grid, and High-Altitude Weather Balloon missions.
- Aiding efforts in day to day lab management, resource allocation, and boosted program reputation and outreach.
- Represented UL Lafayette' EECE department at the state capital to secure six figures in funding for the CAPE program.

FIRST FRC 3616 - *Phenomena Robotics*, Lafayette, LA

Aug 2019 - Aug 2023

➤ **Lead Programmer, Robot Technician, Competition Driver**

- Responsible for developing code and PID tuning a robotic two joint arm.

➤ **Project Manager, Lead Electrician** | May 2022 - August 2023

- Attended the World Championship under my management, robot driving, and robot repair leadership by winning the "most prestigious award in FIRST Robotics" the **Impact Award**.
- Conducted project launch and agile development meetings to define project objectives and scope.
- Set the highest standard of electrical management for team 3616 that is now used to teach proper wiring techniques and has been studied for future robots. Drastically decreased robot wiring time **from 2 weeks to 24 hours**.

Technical Skills

Programming Languages: SQL, HTML, CSS, Python, JavaScript, Java, Arduino, C++, C, AVR & MIPS Assembly, LaTeX, VHDL

Software: GIT, Jquery, Oracle 8, WSL, and Ubuntu Linux.

Hardware: Arduino, ESP32, STM32, MIPS, AVR, CAN, SPI, I2C, UART, RF Communication (UHF, VHF)

Tools: Fusion 360, Altium Designer, Bitbucket, Source Tree, Jira, GitHub, GitLab, Slack, Microsoft Suite, Cura, Putty

Hands on Skills: Soldering, Electronics Prototyping, MultiMeter, Oscilloscope, Power Supply, Software Defined Radios, Spectrum Analyser

Other: Leadership, Agile Development, Scrum

Languages: English (Native), Gujarati (Native)